

Performance Analysis

High Performance Computing

Master in Applied Artificial Intelligence

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Motivation for Performance

- ▶ The goal of parallel programming is to reduce computation time, with the (same) result.
- ▶ However, parallel or concurrent execution must always guarantee the correctness of the result.
- ▶ Faster execution of a program that does not give the correct result is useless!
- ▶ **Latency** is the time it takes for an application to complete its calculations.

Performance Measures: Speedup

- ▶ The speedup value defines the ratio between the sequential version and the parallel version with P processors.

$$\text{speedup} = S_P = \frac{T_1}{T_P}$$

- ▶ T1 represents the time of the sequential version.
- ▶ T_P represents the time of the parallel version.
- ▶ T_{P1}, represents the time of the parallel version, with only 1 processor.

Performance Measures: Efficiency

- ▶ Efficiency measures the ability to use hardware:

$$\text{efficiency} = \frac{S_P}{P} = \frac{T_1}{PT_P}.$$

- ▶ An algorithm is said to have a **linear gain** (linear speedup) when it has an efficiency of 100%.
- ▶ Ideally, 100% efficiency indicates that each new unit of hardware is able to reduce latency proportionally.
- ▶ In practice, the gain is usually sub-linear.

Amdahl Law

Based on a sequential program,

- ▶ What will be the predicted performance gain, based on a parallel computer ?
- ▶ How to calculate/estimate this gain ?

Amdahl Law

$$W_{\text{ser}} = f T_1,$$

$$W_{\text{par}} = (1 - f) T_1.$$

- ▶ Any program is composed of two fractions of instructions: sequential and parallel.
- ▶ The sequential fraction (f) represents the set of instructions that depend on the execution of the previous ones to be executed.
- ▶ The parallel fraction ($1-f$) represents the set of instructions that do not depend on each other to be executed; and that can be executed in parallel.

Amdahl Law

- ▶ T_1 calculation does not change;
- ▶ T_P calculation takes into account the number of processors to execute the parallel fraction.

$$T_1 = W_{\text{ser}} + W_{\text{par}},$$

$$T_P \geq W_{\text{ser}} + W_{\text{par}}/P.$$

$$S_P \leq \frac{1}{f + (1-f)/P}.$$

- ▶ Gain will be at most S_P , that corresponds to the best use for P processors.

Amdahl Law

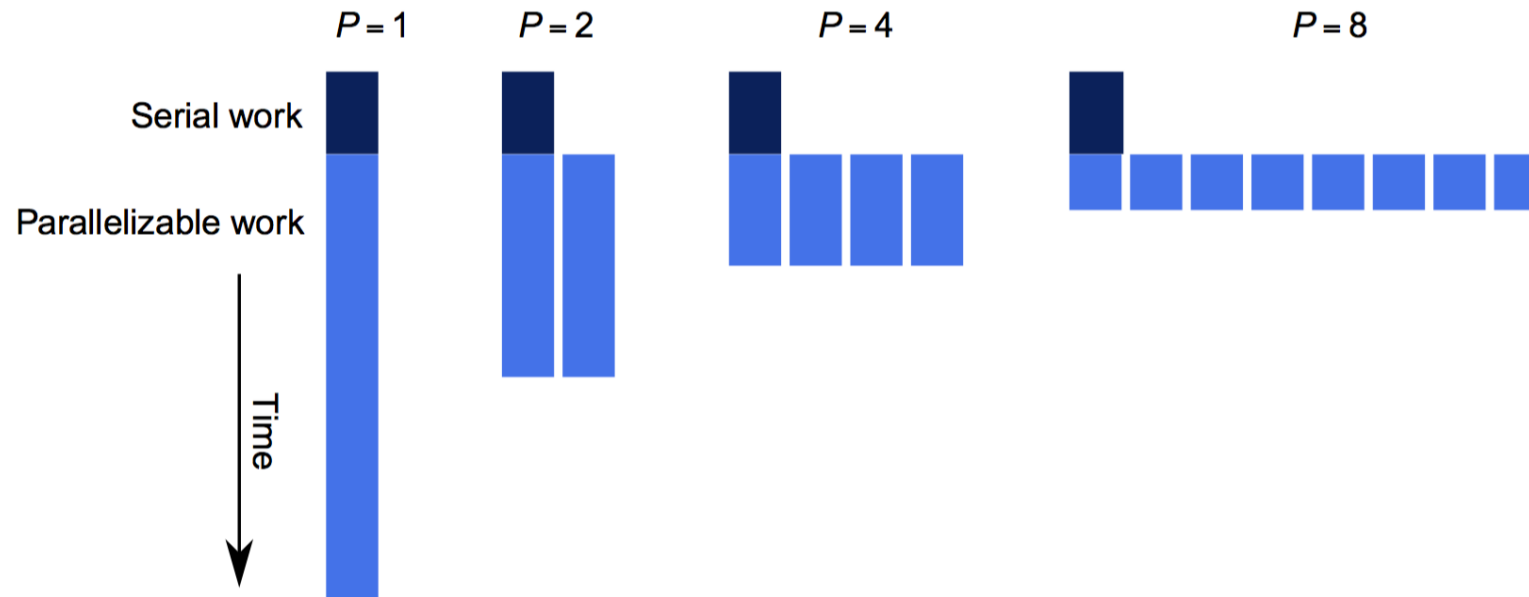


FIGURE 2.4

Amdahl's Law. Speedup is limited by the non-parallelizable serial portion of the work.

Source: McCool, Robison, Reinders, *Structured Parallel Programming*.

Amdahl Law

- ▶ The following expression defines Amdahl's law, which is the maximum possible gain in the execution of a program, for F the sequential fraction, and N the number of cores:

$$Speedup \leq \frac{1}{F + \frac{1 - F}{N}}$$

- ▶ For $F=1$, we have a completely sequential program, which gives us a gain of 1x, identical to a sequential execution.
- ▶ For $F=0.5$, we have a half-sequential and half-parallel program, which gives for infinite cores (N), a maximum gain of 2x.
- ▶ For $F=0$, we have a completely parallel program, whose gain depends linearly on N .
- ▶ The execution gain depends more on F than on N !